

Tether

A quiet daily ritual you share with one person.

Complete build log — everything made, and why.

What this is

Tether is a native iOS app for couples, built local-first and offline. Everything is stored on device and encrypted; there is no server holding anyone's writing. This document records what was built, the decisions behind it, the bugs found along the way, and the reasoning that is easy to lose once the code is written.

At a glance

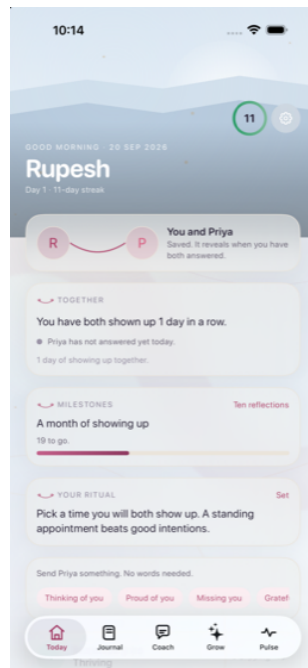
Measure	Value	Note
Swift source files	58	plus a widget target
Lines of Swift	17,123	hand-written
Commits	121	one per logical change
Localised strings	563	English, Hindi, Spanish
Scenes (themes)	11	2 free + glass, rest paid
Wisdom paths	4	Secular, Vedic, Quranic, Biblical
Daily prompts	360	90 per path

1. The app, screen by screen

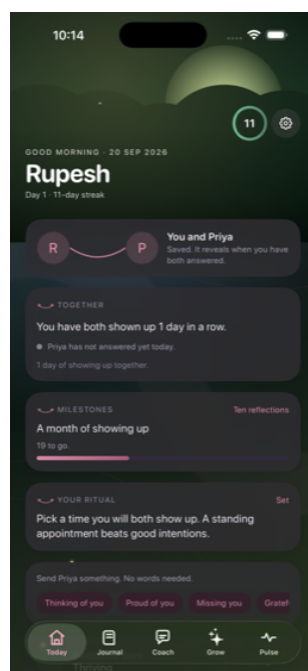
Home — the sky, the tether, the ritual

Home opens on a drawn sky that belongs to the chosen scene, with the day's ritual directly beneath it: a mood, a line of writing, and an explicit choice about who it is for. Below that sit the warmth chips, the pulse, and a two-column gallery rather than a list of buttons.

- **The masthead** is scene art, not a photo. Fireflies at Night, butterflies in the Garden, a seed on the wind in the Meadow.
- **The tether** is drawn as a real slack curve between two people, with their photos. Unpaired, it hangs loose and dashed — visibly waiting rather than broken.
- **The gallery** replaced ten stacked buttons. Each tile carries a live badge and a detail line that changes with state.



Home in the Glass theme — translucent cards over a pale sky, with the journal preview strip showing real thumbnails.



Home in Jungle — canopy silhouette, green throughout, tinted cards. The whole screen is one place.

Journal — yours, and shared

Two tabs that never overlap: **Only me** and **Shared**. Each entry shows its date, its time, and in words whether anyone else can see it. Long-press or swipe from either edge to move an entry between them, or delete it.

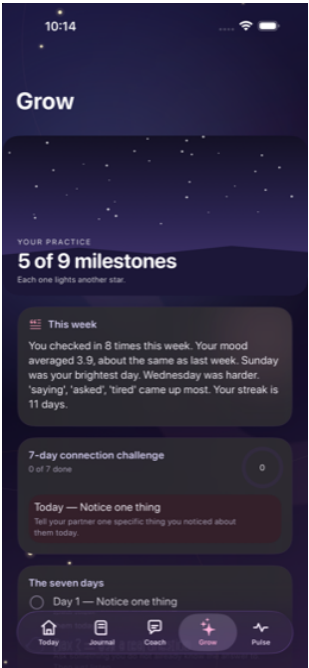
- **Private by default.** Sharing is a deliberate act, not the absence of one.
- **Writable from the journal.** A compose button, which did not exist before this session.
- **Voice, photo or text,** all shown as themselves in the preview strip.



The journal in Jungle — entries grouped by month, each labelled Only me or Shared in words.

Grow — the seven day practice

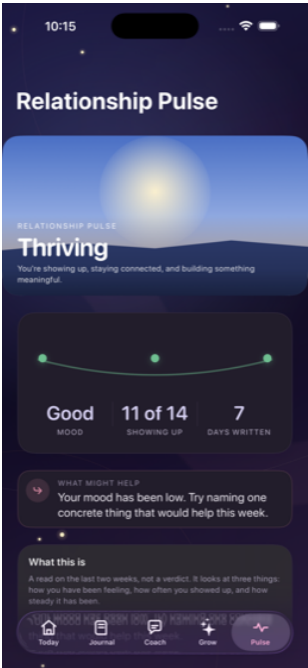
A seven day sequence that builds the habit. Each day closes with a reflection, and completing the week is marked rather than merely counted.



Grow — the seven day practice, with progress and today's step.

Insights — the shape of a stretch of time

Mood over time, distribution, streaks, and a love-language reflection. All computed on device from entries that are already there; nothing is sent anywhere to produce it.



Insights — the pulse, trends and the sentence that says what they actually mean.

2. What was built in this session

Everything below was made in a single working session, in response to reported bugs and to choices you made about direction.

The Two Versions — the feature you picked

Both people write their own memory of the same day. Neither sees the other's until both have written. Then both are shown, side by side, with no verdict.

It only surfaces on days where both wrote *and* at least one found it hard. The single line of commentary is: *"You were both having a hard day, and neither of you knew the other was too. Neither of you is wrong."* It never says who was right. It is a window, not a verdict.

Quiet Week

One tap: *"I need a quiet week."* Three, seven or fourteen days, with an optional line the partner sees. Notifications stop for **both** people. No streak breaks, nothing to catch up on.

The partner's screen says the thing that matters: *"This is not about you. They asked for space, and told you so — which is the opposite of disappearing."*

Every other couples app is built to maximise engagement, which is exactly why none of them has this. A pause button works against the business model. This turns a withdrawal into a message.

Eleven scenes, each with its own horizon

Match phone · Glass · Day · Dawn · Night · Garden · Jungle · Ocean · Dune · Aurora · Meadow.

A scene owns the whole header: its sky, its light, its colours, and its **silhouette**. That last part matters most — recolouring the same two hills green still reads as hills. So Jungle and Garden get layered foliage, Ocean a flat banded waterline, Dune soft shoulders, Meadow low rolling grass, and the sky scenes keep the ridges.

Stars now follow the scene rather than the clock. A dark Jungle sky used to sprout stars at 2am that were not there.

Privacy — the core promise

- Every entry is **encrypted on device** before it is stored, with the key in the Secure Enclave.
- **Private by default.** This was a real bug: entries defaulted to shared, so the Only me tab was always empty and keeping something to yourself was impossible.
- **Face ID** on cold launch and on return from background.
- **Unsent notes** — a place to think before speaking. The partner never learns how many exist.
- Crisis content is detected on device, handled with care, and never distilled into retrievable memory.

3. Bugs found and fixed

Most of these came from you using the app on a real phone, which is the only way they were ever going to surface.

The journal could not be written privately

Found because the Only me tab was always empty.

```
self.visibilityRaw = Visibility.shared.rawValue
```

Every entry defaulted to shared, and nothing ever changed it. There was no control to change it with. Keeping something private meant not writing it at all. Fixed to private by default, with an explicit picker.

Face ID only worked from the app switcher

Found because quitting and reopening opened straight into the journal.

`isLocked` starts `false`, and `lock()` was only called from `scenePhase == .background`. A cold start asked for nothing. Now locked in `init()` as well.

Writing in Shared did nothing

Two causes, both real.

- The visibility picker was **hidden entirely** unless a partner existed, so an unpaired person wrote into Shared and the entry silently saved as private.
- Even when it saved correctly, the journal stayed on the tab you were already on, so the screen looked unchanged.

The phone getting hot

The animated backdrop ran at the display's native rate — 120 times a second on ProMotion — on every screen, forever, redrawing a canvas full of radial gradients. Nothing there moves fast enough for anyone to tell the difference, so the rate is now matched to the content:

Surface	Was	Now
Ambient backdrop	120 fps	8 fps
Header	120 fps	8 fps
Jar	120 fps	16 fps
Drawn note	120 fps	20 fps

“Invalid frame dimension”

This is layout maths producing a NaN or a negative, not a performance problem — so raising the frame rate would only have logged it more often. Three separate sources were found:

- **A NaN clamp that does not clamp.** `min(max(Double(rms) * 14, 0), 1)` returns NaN, because every comparison against NaN is false in Swift. One bad audio sample became a NaN frame width.
- **Unguarded geometry.** A `GeometryReader` inside a scrolling List can be handed a negative size while rows are recycled. Every reader feeding a frame or a shape is now clamped.
- **A division inside a frame.** `count / maxCount` with a zero denominator. Now guarded.

A Spacer was being made into a bar button

`ToolbarItemGroup` makes every child its own bar item, so a `Spacer()` became a zero-width button that UIKit could not satisfy next to the real one. Five keyboard toolbars were affected. They now use a single `ToolbarItem`.

Home grew without bound

Home rendered up to ten full journal cards inline, so the screen got longer every time you wrote — and each card blends a material over the backdrop, which is the most expensive thing on the screen. Replaced with one row that opens the journal, plus a thumbnail strip.

4. Platform work

Audited against the iOS essentials rather than guessing. Present: widget, Reduced Motion, ShareLink, Face ID, haptics, context menus, VoiceOver, Dynamic Type, dark mode, three languages, empty states, StoreKit.

Added this session:

- **Swipe actions** on journal rows, from both edges.
- **Home Screen quick actions** — long-press the icon for New entry, Today's question, or Quiet week.
- **Photo cropping** with pinch, drag and reset.
- **Profile photos**, downsampled before storage.
- **Three accents** — Classic indigo, Bold steel blue, Warm rose — with gender choosing the starting one.

A note on the widget

The widget exists as its own target, `TetherWidget/`. Adding more widget families — a graph, a streak — means writing a timeline provider for each, and paywalling them means sharing entitlement state across targets through an App Group. Both are real work and both remain open.

Localisation

563 strings across English, Hindi and Spanish. A checker script (`check_localization.py`) runs against the build output. Three blind spots were found in it this session, each of which had hidden real gaps:

- It cannot see strings inside **ternary branches**.
- It cannot see strings from **computed `LocalizedStringKey` properties**.
- It cannot see strings built with **interpolation**.

A clean scan is not proof of a complete translation. Roughly forty strings were added by hand after the checker reported nothing wrong.

5. What is still open

Stated honestly, so nothing here reads as finished when it is not.

Item	State	What it needs
Insights in widgets	Not started	Widget target + App Group
More widget families	Not started	A timeline provider each
Paywalled widgets	Not started	Entitlement sharing
Grow after day 7	Open	A design decision from you
Ritual 'everyday'	Open	Clarity on what it should mean
Pulse enrichment	Blocked	A screenshot of what looks empty
Live Activity	Not started	ActivityKit
Spotlight & Siri	Not started	App Intents

How to verify any of this on a device

- Long-press the Tether icon — three shortcuts should appear and work.
- Swipe a journal entry from either edge — Delete and Share.
- Write in Shared, then check it lands in Shared and not Only me.
- Force-quit and reopen — Face ID should ask.
- Scroll Settings and the journal — the console should stay clean.

Built with SwiftUI and SwiftData. Local-first, offline, encrypted, no account. 58 source files, 17,123 lines, 121 commits.